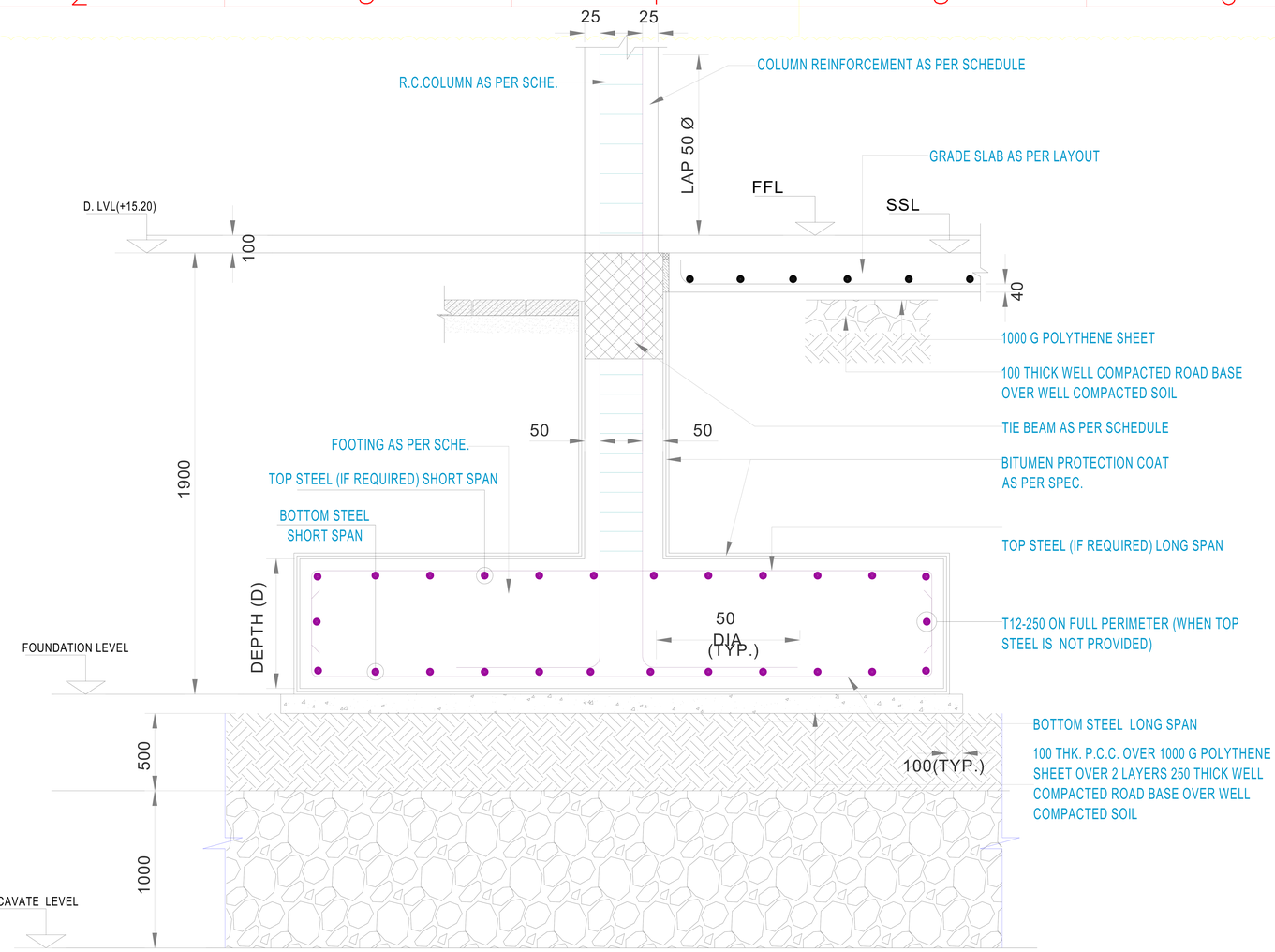

(TYPICAL SECTION FOR FOOTING)

ALLOWABLE BEARING CAPACITY FOR ISOLATED/ STRIP FOOTINGS

Width of Footing	Allowable Bearing Pressure	Modulus of sub-grade
Up to 2.00m	130 kN/m2	15,600 kN/m3
Up to 3.00m	120 kN/m2	14,400 kN/m3
Up to 4.00m	110 kN/m2	13,200 kN/m3
Up to 5.00m	100 kN/m2	12,000 kN/m3

- Excavate and level the existing soil at the Level of -3.50m below the D.LVL (15.20) level established on the site.
- Compact and Level using heavy rollers , the soil at the bottom of excavation at level of -3.50m from the D.LVL established on the site, to at least 95% of its maximum dry density at its optimum moisture content obtained from modified proctor compaction test.
- Reinforce the base of excavation by using crushed rock of size 100 to 150 mm. The rock filling shall be compacted by suitable vibratory roller until no further penetration of rock filing the sand layer during compaction is achieved. The rock fills to be extended to 1 m above the excavation level.
- Place Two (2) layers of selected granular backfill materials (Road Base) 25cm thick each to reach the required foundation level. Each layer shall be compacted by heavy vibratory roller to a degree of compaction not less than 95% of the maximum dry density as obtained by modified proctor compaction test (BS-1377-2:2022).
- Plate bearing test shall be carried out at foundation level as quality control measure to verify the required allowable bearing pressure and total settlement criteria under foundations.
- A 100 mm thick layer of blinding concrete (PCC) should be placed to protect the final compacted surface.
-Shallow foundation can be placed on engineered fill. (i.e., Approximately -1.90m below the D.LVL (15.20) established on the site.)

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

10 February 2026

Building Permit #: PRMT-25-0004677

Appoved By:

STRUCTURAL NOTES	CONCRETE COVERS :
ACI 318-11	Foundations cover = 75mm
F'C/FCU = 28/35 N/mm2	Columns cover = 40mm
FY = 460N/mm2	Walls cover = 30mm
FYt = 420N/mm2	Beams cover = 40mm
	Slabs & Stairs cover = 25mm

Savana
Engineering Consultancy & Interior design

Client	OMAR ABDULLA HASAN ALI ALMANSOORI		
Project	SINGLE STOREY VILLA		
Drg.Title	DETAILS FOUNDATION		
Area	AWAFI	Floortype	
Case ID.	3169896	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406132194
Bldg. No.	BLD-25-01678	Floor No.	
Date	02-2026	Sht. No.	S01
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN

DETAILS FONDATION